

SYST 205: Systems Engineering Principles – Fall 2025
FINAL SYSML & MBSE ASSIGNMENT – GMU SHUTTLE APP (150 points)

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PART 1: Requirements Modeling

1a) Requirements (Shall-Format)

Functional Requirements:

1. REQ-F-01 (Timetable Access): The platform shall render route timetables encompassing each of the five designated transit corridors.
2. REQ-F-02 (Position Monitoring): The platform shall capture and project vehicle whereabouts onto a geographical interface.
3. REQ-F-03 (Tardiness Warning): The platform shall broadcast notifications whenever vehicle tardiness surpasses a five-minute threshold.
4. REQ-F-04 (Spontaneous Booking): The platform shall facilitate rider-initiated transportation requests designating embarkation and disembarkation coordinates.

Performance Requirements:

5. REQ-P-01 (Coordinate Cycling): The platform shall regenerate satellite positioning data at three-second cadences.
6. REQ-P-02 (Processing Velocity): The platform shall derive arrival projections within a two-second envelope.
7. REQ-P-03 (Geolocation Fidelity): The platform shall sustain positional precision not exceeding ten-meter variance.

Security Requirements:

8. REQ-S-01 (Credential Validation): The platform shall confirm user legitimacy through institutional NetID protocols.
9. REQ-S-02 (Channel Protection): The platform shall safeguard network exchanges employing TLS 1.3 cryptographic standards.
10. REQ-S-03 (Privilege Segregation): The platform shall confine backend management capabilities to credentialed operators exclusively.

Usability Requirements:

11. REQ-U-01 (Corridor Isolation): The platform shall empower riders to narrow map renderings to vehicles servicing a designated corridor.
12. REQ-U-02 (Assistive Compatibility): The platform shall accommodate vision-impaired individuals via auditory navigation aids.

Resilience Requirement:

13. REQ-R-01 (Disconnected Operation): The platform shall preserve timetable records on-device to sustain core functionality amid connectivity disruptions.

1b) SysML Requirements Diagram

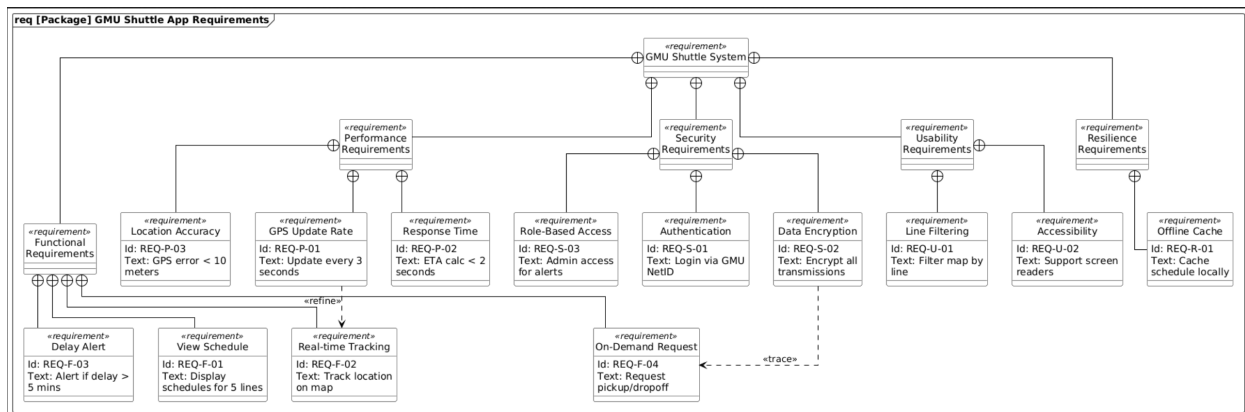


Figure 1: SysML Requirements Diagram (req)

PART 2: Structural Modeling

2a) Block Definition Diagram

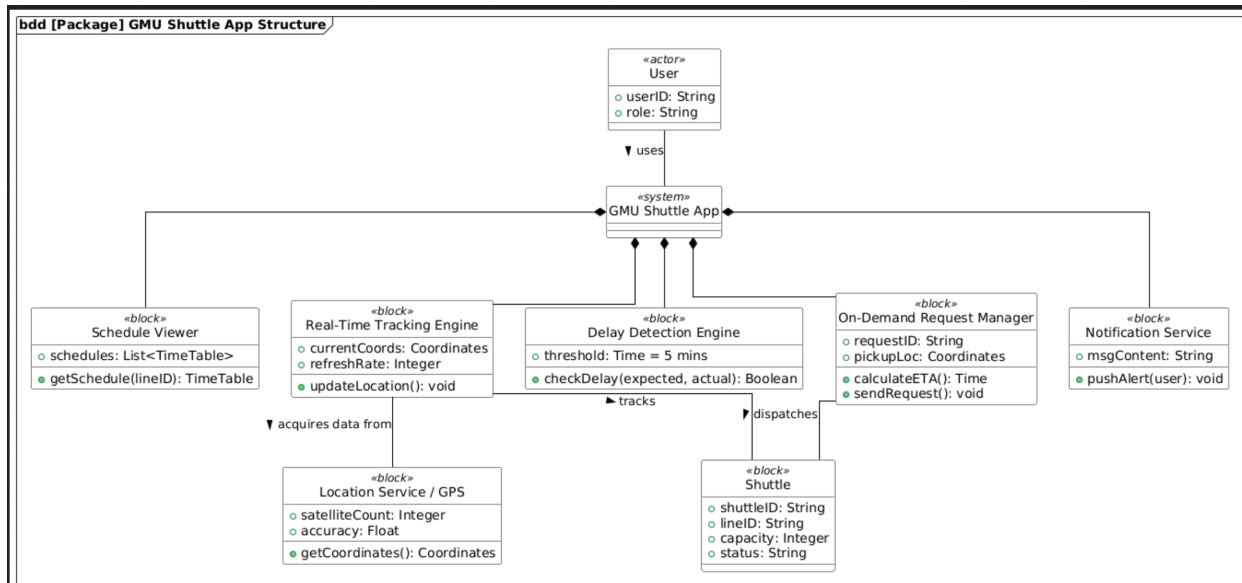


Figure 2: Block Definition Diagram (bdd)

This architectural blueprint delineates how constituent modules interconnect within the campus transit platform. The central hub orchestrates six specialized engines: a Timetable Repository governing scheduling artifacts, a Geospatial Tracker processing instantaneous vehicle bearings, a Punctuality Analyzer flagging temporal discrepancies, a Booking Coordinator managing rider solicitations, an Alert Dispatcher broadcasting user advisories, and a Satellite Interface harvesting positional telemetry. An external Commuter persona engages the ecosystem, while the Vehicle entity encapsulates tangible fleet assets carrying identifiers for unit designation, corridor assignment, occupant threshold, and operational disposition. Solid rhombus connectors signify containment bonds wherein subsidiary modules constitute indispensable fragments of the overarching construct, whereas labeled vectors demarcate intelligence pathways linking discrete components.

PART 3: Behavioral Modeling

3a) Activity Diagram

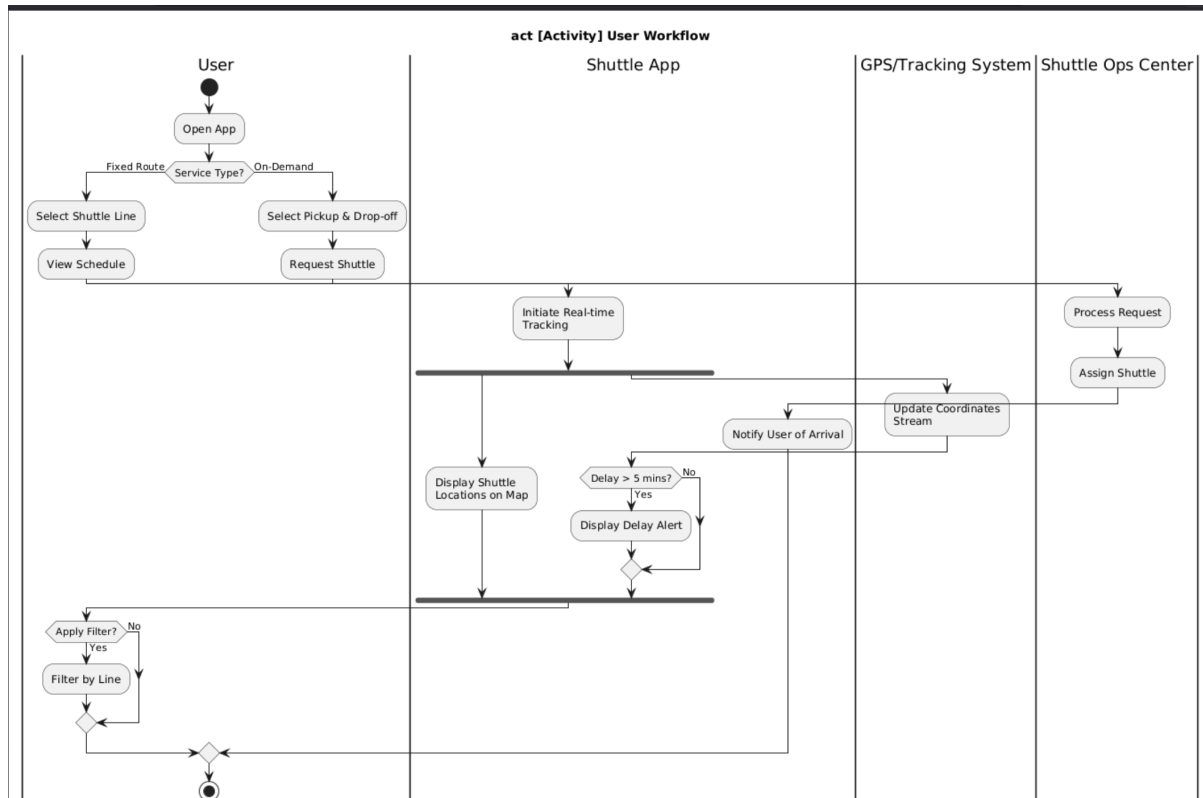


Figure 3: Activity Diagram (act)

This procedural schematic traces rider engagement sequences partitioned across four accountability lanes: Commuter, Transit Platform, Positioning Infrastructure, and Dispatch Headquarters. The choreography commences upon application launch whereupon a branching juncture (Mode Selection?) bifurcates into Scheduled Service versus Impromptu Transit pathways. Scheduled Service channels riders toward corridor designation followed by timetable consultation. Impromptu Transit directs them through waypoint specification culminating in vehicle solicitation. Horizontal synchronization markers denote parallel execution threads wherein cartographic rendering proceeds concurrently with satellite coordinate streaming. A conditional checkpoint (Tardiness > 5 mins?) activates warning propagation when punctuality degrades, and a subsequent fork (Narrow Results?) permits corridor-specific viewport refinement.

3b) Sequence Diagram

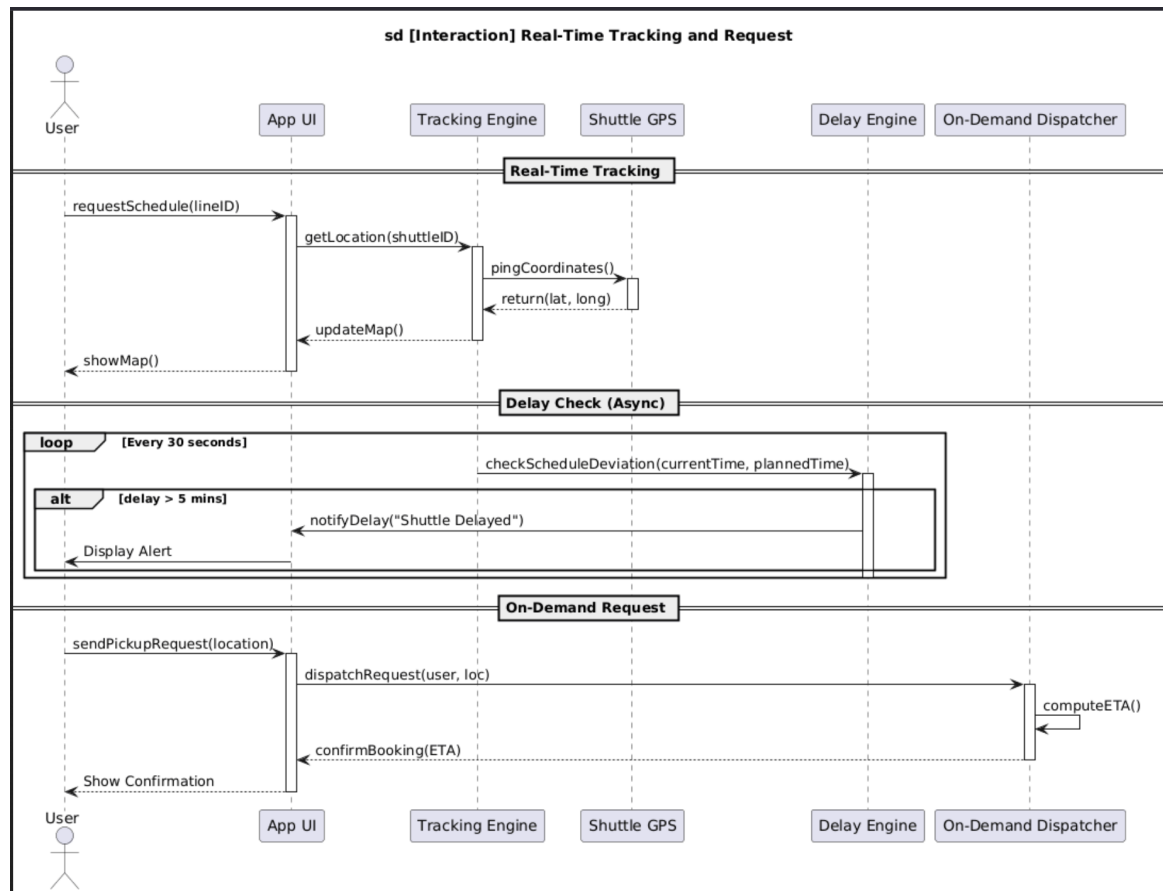


Figure 4: Sequence Diagram (sd)

Temporal Characteristics (Blocking versus Non-Blocking Exchanges):

This chronological blueprint reveals how modules transmit intelligence employing both halting and autonomous dispatch paradigms. Halting invocations such as `fetchTimetable(corridorID)` and `obtainBearing(vehicleID)` utilize continuous arrow glyphs, compelling the originator to suspend execution pending reciprocation indispensable when the viewport necessitates refreshed artifacts prior to canvas reconstruction. Conversely, the Punctuality Surveillance segment executes autonomously through an iterating envelope cycling at thirty-second periods, permitting uninterrupted schedule variance scrutiny without immobilizing the principal interaction thread. Embedded within this iteration, a conditional envelope activates tardiness broadcasts solely when slippage breaches the five-minute ceiling. Responsiveness assumes paramount significance for mobility platforms because sluggish coordinate regeneration or projection computation directly undermines commuter confidence and utility. Stale positioning data risks stranding riders at incorrect waypoints. Hence, the three-second telemetry cadence alongside the two-second projection envelope guarantees the ecosystem delivers punctual, trustworthy intelligence for temporally-constrained journey arbitration.

3c) State Machine Diagram

Modeled Artifact: Booking Transaction Entity

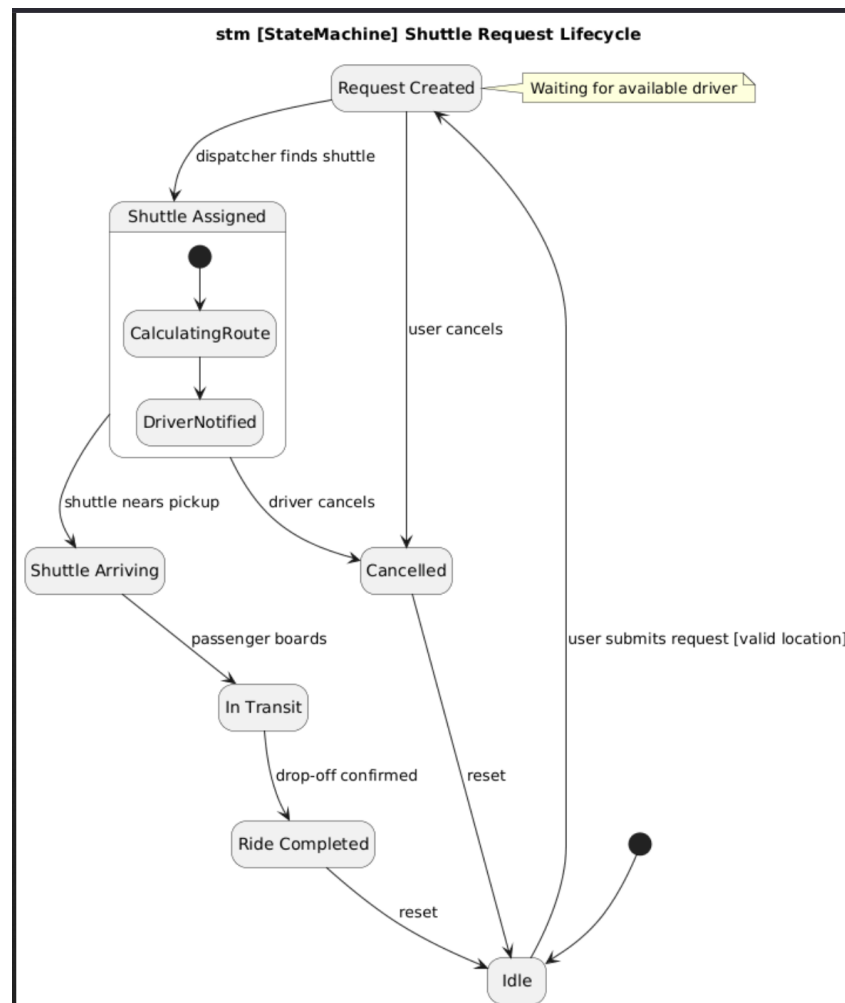


Figure 5: State Machine Diagram (stm)

Analytical Rationale:

Embracing model-centric engineering paradigms substantially alleviates the burden of governing escalating operational intricacy through methodical graphical blueprints encoding condition progressions, stimulus responses, and boundary parameters. When platforms incorporate myriad concurrent threads, anomaly mitigation routines, and human touchpoints, conventional narrative documentation grows unwieldy for upkeep and attestation. The booking transaction lifecycle exemplifies this by articulating seven discrete conditions (Solicitation Lodged, Vehicle Allocated, Vehicle Approaching, Journey Underway, Voyage Concluded, Annulled, and Dormant), a nested condition cluster housing internal phases (Pathfinding and Operator Alerted), boundary stipulations [legitimate waypoint], and diverse stimulus catalysts. Explicit condition articulation proves especially pivotal for mobility and embedded control ecosystems where tangible occurrences demand dependable, foreseeable reactions. Each booking transaction must traverse well-demarcated phases guaranteeing commuters avoid abandonment, operators secure precise directives, and the ecosystem rebounds elegantly from terminations or malfunctions. Absent rigorous condition articulation, fringe scenarios like concurrent rider-operator terminations or connectivity lapses during pathfinding could precipitate erratic ecosystem conduct and diminished commuter satisfaction.

PART 4: Digital Twin Extension

4a) Intelligence Streams for Virtual Replica Assembly

Fabricating a computational doppelganger of the campus mobility network mandates diverse intelligence tributaries spanning instantaneous feeds and archival repositories. Foremost, perpetual satellite telemetry harvesting each vehicle's meridian coordinates, azimuth bearing, and traversal velocity furnishes the bedrock for mirroring corporeal fleet dispositions within the synthetic construct. Secondly, powertrain wellness metrics encompassing propellant reserves, mechanical diagnostics, pneumatic tire inflation, and servicing chronicles would undergird anticipatory analytics for forecasting breakdowns or operational interruptions. Thirdly, ridership flux intelligence from automated boarding tallies, seating occupancy detectors, and credential validation logs would illuminate patronage rhythms across diurnal intervals. Fourthly, arterial congestion and atmospheric feeds from peripheral sources incorporating thoroughfare saturation indices, meteorological readings, and institutional calendar happenings would enrich simulation authenticity. Fifthly, operator conduct telemetry tracking acceleration signatures, dwell intervals, and itinerary conformance would refine emulation precision. Conclusively, longitudinal operational archives documenting punctuality ratios, tardiness frequencies, and cyclical ridership oscillations would nourish algorithmic prediction engines to anticipate forthcoming ecosystem conduct and enhance timetable orchestration.

4b) Emulation Dividends

Computational rehearsals leveraging virtual replicas could markedly bolster transit dependability by empowering administrators to experiment with timetable modifications, itinerary alterations, and fleet dimensioning within a consequence-exempt sandbox preceding campus rollout. Illustratively, planners might emulate ramifications of inaugurating supplementary corridors during congestion peaks or prognosticate how singular vehicle incapacitation cascades across interdependent timetables, thereby enabling preemptive contingency formulation. From a protective standpoint, the synthetic construct could rehearse intrusion scenarios encompassing coordinate falsification or illicit dispatch penetration to expose vulnerabilities and drill countermeasure protocols without imperiling actual commuters or rolling stock.

4c) Hazards and Constraints

A salient peril accompanying virtual replicas in mass mobility contexts involves safeguarding confidential intelligence. The granular telemetry and ridership records indispensable for constructing a veracious synthetic construct spanning trajectory surveillance, patronage rhythms, and potentially distinguishable student particulars constitutes an enticing quarry for nefarious entities. Unauthorized acquisition of such repositories could divulge personal commuting habits, facilitate harassment, or permit adversaries to camouflage tangible interference by corrupting the emulation. Moreover, excessive reliance on virtual replica projections might induce administrators to defer excessively to computational forecasts without cross-referencing corporeal observations, thereby risking oversight of authentic operational aberrations eluding the construct's presuppositions.

PART 5: Reflection

5a) Supreme Modeling Obstacle

Amongst all schematics fabricated, the specification blueprint posed the steepest ascent owing to the imperative for deliberate traceability linkages traversing heterogeneous mandate classifications. Whereas operational schematics adhere to comparatively straightforward sequential or conditional reasoning, the specification blueprint necessitated methodical contemplation regarding how apex-tier ecosystem mandates fragment into granular functional, throughput, protective, ergonomic, and robustness stipulations. Crafting purposeful «derive» and «trace» bonds exemplified by coupling Coordinate Cycling throughput mandates to Position Monitoring functional mandates for implementation dependency, or associating Channel Protection with Spontaneous Booking for protective coverage demanded comprehending inter-mandate kinships transcending mere enumeration. Furthermore, preserving schematic legibility whilst accommodating thirteen annotated mandates bearing identifiers, prose elaborations, and relational vectors presented formidable compositional hurdles.

5b) How Model-Centric Tenets Enriched Platform Comprehension

Foundational model-centric tenets particularly traceability, lifecycle cognizance, and the demarcation between architectural versus operational vantages materially deepened my grasp of the transit platform as a functioning tangible ecosystem. Traceability illuminated that every platform capability exists to gratify documented mandates, and that mandate modifications permeate throughout the entire design tapestry. Exemplifying this, connecting the Channel Protection mandate to the Spontaneous Booking capability clarified why encrypted conduits matter when conveying rider whereabouts. Lifecycle cognizance reframed the transit platform not as a crystallized artifact but as an evolving ecosystem progressing through mandate harvesting, architectural drafting, implementation, attestation, and perpetual stewardship each juncture demanding distinct documentary and verification regimens. Distinguishing architectural vantages (the structural blueprint depicting modules like Geospatial Tracker and Alert Dispatcher) from operational vantages (procedural and chronological schematics revealing temporal interplays) underscored that ecosystem mastery demands dual perspectives: cataloging constituents proves inadequate without grasping their collaborative choreography. The condition progression schematic particularly embodied lifecycle thinking by tracing a solitary artifact (Booking Transaction) through its complete tenure from inception to culmination. This poly-perspective methodology parallels professional engineering squadron organization, wherein system architects concentrate on architecture whilst implementers focus on runtime choreography.

5c) SysML Deficiencies for Handheld Platform Modeling

A conspicuous SysML deficiency when representing handheld software concerns its absence of indigenous constructs for viewport and experiential orchestration. Smartphone platforms hinge heavily upon navigation choreographies, tactile gestures, kinetic embellishments, and adaptive arrangements that directly sculpt patron contentment, yet the notation proffers no canonized schematic varieties for depicting wireframes, viewport taxonomies, or gesture vocabularies. Although procedural schematics can abstract patron journeys at conceptual altitudes, they cannot articulate the visual and tactile subtleties differentiating pedestrian platforms from exceptional ones. Consequently, squadrons employing SysML for handheld endeavors frequently necessitate supplementary prototyping or mockup utilities, introducing fragmentation between engineering blueprints and authentic viewport specifications.

5d) Juxtaposing Narrative-Centric and Model-Centric Methodologies

Executing this undertaking illuminated stark divergences between conventional narrative-driven workflows and model-centric engineering. Under a narrative regime, I would have drafted disparate prose documents narrating mandates, ecosystem architecture, and operational choreographies in paragraph form conceivably accompanied by rudimentary illustrations. While habituated and demanding no specialized instrumentation, this methodology harbors notable frailties: mandates entombed within flowing prose resist tracing, cross-document contradictions evade detection, and revising one passage neglects flagging dependent passages warranting amendment. The model-centric alternative, despite demanding preliminary investment in mastering notation conventions and tooling, delivers compelling dividends. The specification blueprint rendered traceability explicit and graphical. I could instantaneously discern which throughput stipulations buttressed which functional capabilities. The architectural blueprint compelled rigorous interface and containment definitions rather than nebulous prose characterizations. Operational schematics (procedural, chronological, condition progression) guaranteed contemplation of timing, parallelism, and boundary scenarios that narrative renditions routinely neglect. Beyond these dividends, natural language's intrinsic ambiguity frequently spawns divergent interpretations under narrative workflows, whereas the notation's rigorous grammar mandates clarity guaranteeing that technical parameters like 'geolocation fidelity' carry identical connotations for architects and implementers alike. Additionally, model-centric engineering institutes a unified authoritative repository; modifying an architectural attribute (say, Vehicle Occupant Threshold) instantaneously propagates across all linked architectural and operational artifacts. Such automated synchronization eradicates versioning conflicts and datum inconsistencies plaguing fragmented narrative repositories as specifications metamorphose. Nevertheless, model-centric methodologies harbor drawbacks: the notation's learning trajectory proves precipitous, tooling malfunctions (like the Innoslate persistence failure I encountered) can derail momentum, and evaluators unfamiliar with the notation may struggle to scrutinize outputs. For this particular undertaking, the model-centric pathway proved unambiguously advantageous because it enforced discipline and surfaced conceptual voids that would have lurked concealed within conventional prose. For less elaborate initiatives or squadrons lacking modeling proficiency, narrative-centric methodologies might remain more pragmatic.

Software Requirements

Preliminary schematic fabrication employed Innoslate as the designated modeling environment. Regrettably, recurring persistence malfunctions within Innoslate obstructed dependable artifact conservation, necessitating migration to PlantUML as a substitute notation-capable environment. PlantUML functions as a freely accessible, community-maintained, script-driven modeling utility accommodating specification schematics (req), architectural blueprints (bdd), chronological schematics (sd), condition progression schematics (stm), and procedural schematics (act). Its script-oriented workflow facilitates versioning oversight, reproducibility, and documentary integration. Every schematic incorporates notation-compliant frame annotations per assignment stipulations.

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